

3(a)	all three correct ; with the non-competitive inhibitor Z with the competitive inhibitor Y without any inhibitor X	1
3(b)	four from: V_{max} 1 X and Y same V_{max} of 10 au ; 2 V_{max} of, X/Y, higher than Z/ORA ; A (V_{max} of), X/Y, 10 au v Z 5 au A (V_{max} of), X/Y, double the V_{max} of Z K_m 3 X and Z same K_m ; A K_m of both is 4 mmol dm⁻³ 4 X/Z, lower K_m than Y/ORA ; A K_m of, X/Z, 4 mmol dm⁻³ v Y 6.5 mmol dm⁻³ 5 reference to affinity for substrate ;	4
3(c)	four from: 1 double helix ; 2 strands are held together by hydrogen bonds (between bases) ; 3 complementary base pairing / described as A-T and C-G ; A purine pairs with pyrimidine R thiamine 4 antiparallel stands / strands are 3' to 5' and 5' to 3' ; A strands run in opposite directions 5 (each strand has a sugar phosphate backbone with) phosphodiester bonds ; 6 (monomers / units / DNA) are (DNA) nucleotides / polynucleotide strands ; 7 (nucleotide =) <u>deoxyribose</u> sugar, phosphate, nitrogenous (organic) base ; A points from a diagram	4

Question	Answer	Marks
3(d)	<i>two from:</i> 1 <i>idea that</i>, hydrogen peroxide, damage / breaks, DNA <u>and</u> repair errors (may) occur ; 2 (so leads to) incorrect, nucleotide / base, inserted (during replication) / change in, nucleotide / base, sequence (of DNA / RNA) ; 3 new allele (may be) formed ; 4 may result in an altered polypeptide / AW ;	2